Certified causal bounds for an auxiliary Einstein–scalar response

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Abstract

For one previously registered quadrupole in a conditional five-dimensional Einstein–scalar material-shell model, we enclose an evaluated component of the auxiliary quadratic forced response. Twenty exact-area source cells and a bound for every retained feedback iteration imply $1.7 \times 10^7 < \langle W_{yy} \rangle < 6.5 \times 10^{10}$ throughout $|x - 1/4|, |r - 5/4| \le 10^{-6}$. All 23 bulk amplitudes are retained. The domain of dependence stays away from the shell; its contributions vanish by causality. The evaluated point also gives $\|W\|_X > 8.8 \times 10^6$, below the retained upper bound 5.228×10^{13}. A four-term true-operator Neumann expression has truncation tail below 1035, but remains unevaluated. The methods are standard interval-integral and norm estimates applied to the retained model-specific certificates. The result is an auxiliary, frame/gauge-dependent response enclosure, not a global waveform, standalone physical constraint solution, nonlinear spacetime, or physical-force or cosmological-origin claim.

1 Retained conditional problem

The inputs are the published geometric-source construction and its separate response-attribution calculation [1, 2], pinned to the repository commit recorded in the accompanying reproducibility manifest. We retain their background, first solution, gauge, angular conventions, regularity assumptions, and inverse estimates. This note does not reprove those upstream assumptions or replace the actual first solution by its comparison. The fixed quadrupole is $Y = n_0^2 - n_1^2$ on the unit S^3, with the retained actual master and shell-preserving relative-harmonic gauge field. The scaled coordinates are $x = t/\ell$, $r = z/\ell$, with $\ell = 10^{-6}$ and $dy = -\rho\, dz$. Tensor entries use the retained radially parallel background frame, whose normal vector is $e_y = \partial_y$.

The fixed finite angular state has 23 amplitudes, retaining all metric blocks, the scalar, and both transverse-traceless orientations. Its squared norm $|\cdot|_{\mathcal{H}}^2$ is the probability-normalized S^3 integral of the metric Frobenius square plus the scalar square. On

$$D = \{(x, r) : 0 \le x \le 3,\ r \ge 0,\ x + r \le 8\}, \qquad \|V\|_X = \max\left\{ \sup_D |V|_{\mathcal{H}}/3, \sup_D |V_x|_{\mathcal{H}}, \sup_D |V_r|_{\mathcal{H}} \right\},$$

the accepted reduced equation and complete linear operator are

$$U = U_{\text{free}} + \mathcal{K}F + \mathcal{L}G + TU, \tag{1}$$

$$TV = \mathcal{K}(AV_x + BV_r + D_0 V) + \mathcal{L}(MP_N V|_{r=0}), \qquad \|T\|_{X \to X} \le q = \frac{2109}{10^6} < 1. \tag{2}$$

The source normalization is $F_{\mathrm{metric}} = -2\ell^2\rho^2 Q_{\mathrm{red}}$ and $F_{\mathrm{scalar}} = \ell^2\rho^2 Q_\phi$, for second parameter derivatives. The boundary vector G includes the physical shell rows and normal harmonic source; T includes the complete coupled Robin matrix. In particular,

$$\mathcal{K}f(x,r) = \frac{1}{2}\int_0^x \int_{r-x+s}^{r+x-s} f_{\mathrm{ext}}(s,v)\,dv\,ds, \qquad \mathcal{L}g(x,r) = -\int_0^{\max(x-r,0)} g(s)\,ds,$$

where the extension is the registered odd/even reflection $P_N - P_D$. The accepted flat initial wedge and corner compatibility give $G(0) = 0$ and the required C^1 integral fields with zero Cauchy data.

Set

$$S = \mathcal{K}F + \mathcal{L}G, \qquad W = (I-T)^{-1}S, \qquad U_{\mathrm{hom}} = (I-T)^{-1}U_{\mathrm{free}}.$$

Thus $U = U_{\mathrm{hom}} + W$. This is an auxiliary decomposition of the reduced operator problem. The prescribed second initial data may themselves encode quadratic constraints. Neither summand is asserted separately to satisfy the complete physical constraint and gauge system. Throughout, $\lambda(V) = \langle V_{yy}\rangle_{S^3}$ denotes the probability-sphere mean of the covariant normal-frame component. Its dual $\mathcal{H}$ norm is one, attained by the retained constant pure yy tensor. It commutes with x, r derivatives, and the exact angular projection identifies $\lambda(F)$ with the constant yy forcing coefficient.

2 Causal-cone value calculation

Set $t_0 = 1/4$, $r_0 = 5/4$, and

$$C = \{0 \le x \le t_0,\ |r - r_0| \le t_0 - x\}.$$

Every backward cone from a point of C stays inside C. Moreover $r - x \ge 1$ and $x + r \le 3/2 < 8$ there. Thus the restriction of the actual global auxiliary W satisfies

$$W|_C = \mathcal{K}F|_C + \mathcal{K}(AW_x + BW_r + D_0W)|_C.$$

Both the affine shell lift and the Robin feedback vanish exactly by domain of dependence. The material-shell first solution still determines F; no boundary law or bulk coupling has been changed.

Use the same norm restricted to C. Endpoint differentiation and the retarded integral give

$$\|\mathcal{K}f\|_{X_C} \le t_0\,\|f\|_{C^0(C;\mathcal{H})}, \qquad |\mathcal{K}f(t_0,r_0)|_{\mathcal{H}} \le \frac{t_0^2}{2}\,\|f\|_{C^0(C;\mathcal{H})}.$$

The retained individual full-state coefficient bounds imply

$$\sigma = 10^{-4} + 10^{-4} + 3\times 10^{-6} = \frac{203}{10^6}, \quad q_C = t_0\sigma = \frac{203}{4000000} < 1, \quad \|W\|_{X_C} \le B_C := \frac{t_0 F_{\mathcal{H},C}}{1-q_C}.$$

This follows directly from the restriction inequality for the already specified W; a separate local inverse theorem is unnecessary. All 23 amplitudes enter the source norm and the coefficient majorants. At the apex, the complete feedback allowance is

$$|\lambda(W - \mathcal{K}F)(t_0,r_0)| \le E := \frac{t_0^2}{2}\sigma B_C = \frac{203F_{\mathcal{H},C}}{127993504}.$$

It bounds all feedback iterations together.

The calculation partitions C into 20 exact-area intersections of a uniform rectangular mesh with spacing $h = t_0/4 = 1/16$. In time row $i = 0,\dots,3$, the radial indices are $k = i,\dots,7-i$.

The two end intersections have area $h^2/2$ and every intermediate intersection has area h^2. Their union is C and their interiors are disjoint; the total area is $t_0^2 = 1/16$. Source intervals on the rectangular supersets enclose the actual source on each intersection, including the inherited first-field C^3 uncertainty. These are range-times-area integral bounds, not sampled quadrature errors. In particular,

$$\frac{1}{2}\sum_j |C_j|\, l_j \le \lambda(\mathcal{K}F)(t_0, r_0) \le \frac{1}{2}\sum_j |C_j|\, u_j, \qquad l_j \le \lambda(F)|_{C_j} \le u_j.$$

Negative cell lower endpoints are included in the signed sum. The full-state source cap uses exact probability-sphere Gram matrices, including both mixed-block multiplicities and both tensor orientations. The final accepted cap is $F_{\mathcal{H},C} = 17381837650838$.

Quantity	Safe outward inequality
$\lambda(\mathcal{K}F)(t_0, r_0)$	$5.4 \times 10^7 < \lambda(\mathcal{K}F) < 6.4 \times 10^{10}$
Complete feedback allowance	$E < 2.757 \times 10^7$
$\lambda(W)(t_0, r_0)$	$2.64 \times 10^7 < \lambda(W) < 6.4 \times 10^{10}$

The reproducibility reports retain the exact rational endpoints. The interval is wide; its strict positive lower endpoint is the certified feature.

For a continuous-region consequence, set $\epsilon = 10^{-6}$ and

$$R_\epsilon = \{|x - 1/4| \le \epsilon,\ |r - 5/4| \le \epsilon\}.$$

This entire rectangle lies in the enlarged backward cone with apex $(t_+, r_0) = (t_0 + 2\epsilon, r_0)$, whose minimum $r - x$ is $1 - 2\epsilon > 0$. The accepted global source cap gives

$$B_+ = \frac{t_+\, 17387513156573}{1 - t_+\sigma}, \qquad |\lambda(W)(x, r) - \lambda(W)(t_0, r_0)| \le 2\epsilon B_+.$$

Both mean derivatives are bounded by B_+ and the joining segment stays inside the convex enlarged cone. The allowance is below 8.695×10^6. Consequently, throughout the complete closed rectangle,

$$\boxed{1.7 \times 10^7 < \lambda(W)(x, r) < 6.5 \times 10^{10}.} \tag{3}$$

No derivative bound for F, point-sampling sign inference, or shell-contact approximation enters this extension. The result concerns the covariant yy frame component's probability-sphere mean in the registered gauge. The variables and field normalization are the retained scaled ones, with no SI calibration or physical-force interpretation.

3 Global norm consequences and an accuracy target

The sphere mean obeys $|\lambda(W)(x, r)| \le 3\,\|W\|_X$, because the value term of the registered norm has weight three. The exact evaluated apex lower endpoint L therefore yields

$$\|W\|_X \ge L/3 > 8.8 \times 10^6.$$

This is a stronger global norm lower bound than the independent weak-source witness in Appendix A. It does not evaluate a global coefficient field or identify a physical energy. The retained actual source bounds and Green-map estimates give

$$F_{\mathcal{H}} \le 17387513156573, \quad G_{\mathcal{H}} \le 3851400, \quad \|S\|_X \le 3F_{\mathcal{H}} + G_{\mathcal{H}} = 52162543321119 =: s_*.$$

Therefore $\|W\|_X \le s_*/(1-q) < 5.228 \times 10^{13}$. For $V_N = \sum_{j=0}^{N-1} T^j S$,

$$(I-T)V_N - S = -T^N S, \qquad \|W - V_N\|_X \le \frac{q^N s_*}{1-q}.$$

For $N = 4$ this last bound is strictly below 1035. It excludes numerical input, integration, and reconstruction errors. V_4 here is a true-operator expression, not a computed coefficient field or numerical residual certificate. Its Robin defect is $-MP_N(T^3S)|_{r=0}$, so finite partial sums need not satisfy the exact coupled boundary law. A numerical candidate must account for the complete integral residual against the true retained inputs.

4 Reproducibility and limits

The accompanying package records immutable upstream hashes, exact rational arithmetic, independent internal mathematical review, and isolated execution results. It distinguishes its own checks from the accepted upstream source certification. The weak cutoff and contraction arguments are standard; their application supplies a quantitative conditional bound for this registered state and norm. Internal AI-assisted review does not establish external peer review or originality.

These bounds do not establish a full-domain evaluated second response, a finite-amplitude nonlinear family, a physical force, or a distinctive cosmological prediction. In particular, the norm lower bound includes derivatives and is not a value-only amplitude bound. Any separately certified coordinate component retains its gauge and initial-prescription dependence. No conclusion about a higher-dimensional origin of the observed Big Bang follows from this calculation.

A Independent weak-source lower bound

Write $P = \partial_x^2 - \partial_r^2$. The normalized retarded fundamental solution $\frac{1}{2}\mathbf{1}_{x \ge |r|}$ gives $P\mathcal{K}F = F$ in the physical interior, distributionally. Since $\mathcal{L}G$ depends on $x-r$ alone, $P\mathcal{L}G = 0$, also distributionally. These identities require no derivative estimate for the continuous quadratic source.

Let λ be the sphere mean of the covariant yy metric component. Its dual $\mathcal{H}$ norm is one: its Riesz vector is the retained constant pure yy tensor of value one. Consequently

$$|\lambda(V_x)|,\ |\lambda(V_r)| \le \|V\|_X .$$

The value weight three in X does not enter this derivative estimate. The exact angular projection identifies the constant yy source coefficient with $f_0 = \lambda(F)$; the other scalar harmonics have zero sphere mean.

The published actual-source report encloses $f_0 \ge c_- > 16 \times 10^9$ continuously on the square

$$R = [1/4, 1025/4096] \times [5/4, 5121/4096], \qquad h = 1/4096.$$

The exact rational c_- is read from the hash-authenticated report by the reproducer. The square lies strictly inside D and has $r - x > 0$. In fact its boundary lift vanishes and its backward cones remain at positive depth; ordinary unreflected integration suffices for the local weak identity.

For any nonnegative $\chi \in C_c^\infty(R^\circ)$, one weak integration by parts gives

$$c_- \int \chi \le \int f_0 \chi = -\int \lambda(S_x)\chi_x + \int \lambda(S_r)\chi_r \le \|S\|_X \left(\|\chi_x\|_{L^1} + \|\chi_r\|_{L^1}\right).$$

Take products of smooth one-dimensional plateau cutoffs with compact support inside the two coordinate intervals. Each has total variation two and mass $m_\varepsilon \to h$. The product has mass m_ε^2 and summed first variation $4m_\varepsilon$. Passing to the limit yields

$$\|S\|_X \geq \frac{c_- h}{4} > \frac{(16 \times 10^9)h}{4} = \frac{1953125}{2}.$$

The limit gives a nonstrict inequality. Strictness of the rounded bound comes afterwards from the separately verified slack $c_- > 16 \times 10^9$. Since $S = (I - T)W$, the complete operator bound gives

$$\boxed{\|W\|_X \geq \frac{c_- h}{4(1+q)} > \frac{976562500000}{1002109} > 974000.} \tag{4}$$

The transfer factor is $1 + q$, not the upper-resolvent factor $1 - q$. Equation (4) proves full-state nonvanishing. Coupling prevents inferring a specified response component from this argument alone.

References

[1] R. Maldonado, *Complete geometric quadratic sources and intrinsic shell diagnostics*, research checkpoint and source certificates, https://github.com/maldonado-research/HDblast/tree/d110d99f054ad297fdcc0a7cec07499341bae4b8/research/HDBLAST_GEOMETRIC_BULK_SOURCES_20261016.

[2] R. Maldonado, *Separating the forced increment in the retained formal second response*, auxiliary operator estimate and method note, https://github.com/maldonado-research/HDblast/tree/d110d99f054ad297fdcc0a7cec07499341bae4b8/research/HDBLAST_RESPONSE_ATTRIBUTION_20261016.